ITA: 0501 COMPUTER VISION

CO: 2-AT-1- ANALYTICAL PROBLEMS

B.S. RESHMA
192521342

16/10/26 ITA: 0501 Computer Vision (192521342) B.S. Reshma

1. Thresholding and intensity-level slicing - PCB inspection

a) Binary threshold transformation.

$S = 255, r \geq 128$ o.t : 0

Original	condition	Output (S)
25	$25 < 128$	0
60	$60 < 128$	0
95	$95 < 128$	0
130	$130 \geq 128$	255
165	$165 \geq 128$	255
200	$200 \geq 128$	255
235	$235 \geq 128$	255

An: $S = [0, 0, 0, 255, 255, 255]$

b) Intensity level slicing.

$S = 255, 90 \leq r \leq 170$; o.t = r

Original (r)	condition	output (s)
25	no	25
60	no	60
95	yes	255
130	yes	255
165	yes	255
200	no	200
235	no	235

$S = [25, 60, 255, 255, 255, 200, 235]$

c) Number and percentage of highlighted pixels:

P = highlighted pixel / total pixel $\times 100$

$P = 3/7 \times 100$

$P = 42.86\%$.

d) Mean intensity before and after slicing.

① $\bar{r} = \Sigma r_i / N$

$r = 25 + 60 + 95 + 130 + 165 + 200 / 7 \Rightarrow 130.$

② mean intensity after slicing:

$$\bar{s} = 25+60+255+255+255+20/-1$$

$$\bar{s} = 183.57$$

e) which transformation is more accurate?

An: Intensity - level slicing.

2. Piecewise - linear contrast stretching - x-ray image enhancement:

a) Derivation of slop of each segment:

$$s = \frac{s_a + s_b - s_a}{r_b - r_a}(r - r_a)$$

$$m = \frac{s_b - s_a}{r_b - r_a}$$

Segment 1:

$0 \le r \le 70$: (0,0) and (70,20)

$$m_1 = \frac{20-0}{70-0} = \frac{20}{70}$$

$$m_1 = 0.2857$$

$$s = 2/7\, r.$$

Segment 2:

$70 \le r \le 140$ (140, 230)

$$m_2 = \frac{230-20}{140-70} = \frac{210}{70}$$

$$m_2 = 3$$

$$s = 20 + 3[r - 70)$$

$$s = 3r - 190$$

Segment 3:

$140 < r \le 255$

(140, 230) and (255, 255)

$$m_3 = \frac{255-230}{255-140} = \frac{25}{115}$$

$$m_3 = 0.2174 \; ; \; s = 230 + \frac{5}{23}(r - 140)$$

$$s(r) = \begin{cases} 2/7\, r, & 0 \le r \le 70 \\ 3r - 190 & 70 \le r \le 140 \\ 230 + 5/23(r-140 & 140 \le r \le 255 \end{cases}$$

b) Calculate the transformed value of every pixel.

$$r = [30, 70, 90, 110, 140, 200]$$

① $r = 30$, $s = 2/7\, r$, $2/7(30) = 8.57$

② $r = 70$, $s = 2/7\, r$, $2/7(70) = 20$

③ $r = 90$, $s = 3r - 190$, $270 - 190 = 80$

④ $r = 110$, $s = 3r - 190$, $330 - 190 = 140$

⑤ $r = 140$, $s = 3r - 190$, $= 420 - 190 = 230$

⑥ $r = 200$, $s = 230 + \frac{5}{23}(r - 140)$,

$s = 230 + 5/23(r - 140) = 243.04$

Ans $S = [8.57, 20, 80, 140, 230, 243.04]$

c) Bandwidth and contrast gain:

$W_{ori} = 140 - 70 = 70$

$W_{trans} = 230 - 20 = 210$

Contrast gain factor: $G_1 = \frac{W_{trans}}{W_{ori}} \Rightarrow \frac{210}{70}$

$G_1 = 3$

d) Identify compressed and expanded segments

Segment	Slope	Effect
0-70	0.2857	compresses intensity range
70-140	3	Expands intensity range
140-255	0.2174	compresses intensity range

e) Assessment of diagnostic visibility.

Piecewise-linear contrast stretching enhances details within the specified X-ray region, but it sacrifices some contrast outside that region.

3. Spatial filtering - Impulse noise removal:

$$\begin{bmatrix} 52 & 55 & 58 \\ 54 & 255 & 57 \\ 53 & 56 & 59 \end{bmatrix}$$

$52+55+58+54+57+53+56+59 = 444$

a) Calculate mean filter output:

$I_{mean} = \frac{\text{Sum of 9 pixels}}{9}$

$= \frac{52+55+58+54+255+57+53+56+59}{9}$

$= 77.67$

b) Calculate the weighted-average filter output.

$W = 1/16 \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$.

$I_{weighted} = 1/16 [52 + 2(55) + 58 + 2(54) + 4(255) + 2(57) + 53 + 2(56) + 59]$

$$I_{weighted} = \frac{244 + 444 + 1020}{16}$$

$$= 105.375$$

c) Calculate the median filter output

52, 53, 54, 55, (56), 57, 58, 59, 255

$I_{median} = 56$

d) Calculate the absolute error of each filter:

$$I_{ref} = \frac{444}{8} = 55.5$$

$$E = |I_{fil} - I_{ref}|$$

$$= |77.67 - 55.5| = 22.17$$

$E_{weighted} = |105.375 - 55.5| = 49.875$

$E_{median} = |56 - 55.5| = 0.5$

e) Which filter is most appropriate?

↳ median filter is the most appropriate choice.

---

4. Edge detection - Sobel and Prewitt Gradient Analysis:

a) Calculate G_x and G_y using Sobel.

$G_x = (-1)(10) + (0)(10) + (1)(10) + (-2)(10) + (0)(10) + (2)(90) + (-1)(10) + (0)(90) + (1)(90) \Rightarrow 0 + 160 + 180$

$G_x = 240$

$G_y = (-1)(10) + (-2)(10) + (-1)(1) + (0)(10) + (0)(10) + (0)(90) + (1)(10) + (2)(90) + (1)(90) \Rightarrow -40 + 0 + 280$

$G_y = 240$

b) Gradient magnitude and approximation error:

$$M = \sqrt{G_x^2 + G_y^2}$$

$$= \sqrt{240^2 + 240^2}$$

$$M = 339.41$$

$$M_{approx} = |G_x| + |G_y|$$

$$= 480$$

$$E = \frac{|M_{approx} - M|}{M} \times 100 = \frac{|480 - 339.41|}{339.41} \times 100 = 41.42\%$$

c) calculate gradient direction and edge orientation.

$$\alpha = \tan^{-1}\left(\frac{G_y}{G_x}\right)$$

$$\alpha = \tan^{-1}\left(\frac{240}{240}\right)$$

$$\alpha = 45^\circ$$

d) Repeat using pythagorean equation:

$G_x = 160 \quad G_y = 160$

$$M = \sqrt{160^2 + 160^2} = 226.27$$

Approximation : 320

Percentage error : 41.42%.

e) edge detection:

Sobel : 339.41 > 250 edge detected

Prewitt : 226.27 < 250 no edge

---

5. Global Thresholding and Otsu Segmentation

a) Global mean:

$$m_G = \frac{122}{32}$$

$$m_G = 3.8125$$

b) Iterative threshold:

$$m_1 = \frac{17}{14} = 1.2143$$

$$m_2 = \frac{105}{18} = 5.833$$

$$T_1 = \frac{1.2143 + 5.8333}{2} = 3.5238$$

$$\Delta T = |3.5238 - 3.8125| = 0.2887 < 0.5$$

c) Calculate Otsu's between class variance

$$\sigma_B^2(k) = P_1(k)\, P_2(k)\, [m_1(k) - m_2(k)]$$

$k_4 = 2$: 5.1202

$k =$:

$k = 3 : 5.2506$

$k = 4 : 5.1354$

d) Optimum threshold :

Maximum variance occurs at :

$k^* = 3$

$18/32 \times 100 = 56.25\%$.

e) Compare the thresholds.

Both methods separate levels 0-3 and 4-7 !

The histogram is reasonably suitable for global thresholding